

Advancing Non-Invasive Scoliosis Detection Using Deep Learning

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ABSTRACT

Scoliosis is a complex spinal disorder characterized by an abnormal curvature of the spine, primarily affecting adolescents and often progressing without timely diagnosis [2], [6]. Early detection is crucial to prevent long-term complications [7], but conventional diagnostic methods rely heavily on expert interpretation of radiographs [2], [3], which can be subjective and inaccessible in under-resourced areas [6], [11]. This research proposes a computer-aided diagnostic system using Convolutional Neural Networks (CNNs) to automatically detect scoliosis from grayscale X-ray images. The model was trained on a publicly available dataset comprising 259 labelled X-ray images [1], achieving a test accuracy of 78.85%. Although the dataset size and class imbalance (188 scoliosis vs. 71 normal) limit generalization, results indicate strong feasibility for early detection. The system was integrated into an interactive Streamlit web application, enabling real-time predictions while complementing radiologists' assessments rather than replacing them. The application emphasizes usability and accessibility, particularly for preliminary screening in rural clinics, schools, or telehealth services [11]. Overall, this study demonstrates that deep learning models [4], [5], [9], [10] can offer a scalable, cost-effective, and reliable solution for assisting scoliosis diagnosis in both clinical and non-clinical environments.

Keywords: Scoliosis, Deep Learning, CNN, X-ray classification, Medical Imaging, Streamlit, Healthcare AI

I. INTRODUCTION

Scoliosis is a complex three-dimensional spinal deformity characterized by a lateral curvature and vertebral rotation of the spine, most commonly diagnosed during adolescence. Among the various types, Adolescent Idiopathic Scoliosis (AIS) is the most prevalent, affecting approximately 2–3% of children between the ages of 10 and 16 [6], [11]. If left untreated, scoliosis can progress significantly, leading to chronic back pain, respiratory issues, and impaired quality of life [7].

Conventional diagnostic methods involve clinical assessments and radiographic analysis, typically performed by orthopaedic specialists. Diagnosis is usually confirmed through X-ray imaging, where the Cobb angle is used to quantify the degree of curvature [2]. However, manual diagnosis is time-consuming, expertise-dependent, and prone to human error [3]. Moreover, rural or under-resourced regions often lack trained radiologists, making early detection and regular screening difficult [4].

With the rise of Artificial Intelligence (AI) and machine learning in healthcare, there is growing interest in leveraging computer vision for disease detection and classification [9], [10]. Convolutional Neural Networks (CNNs) have shown exceptional promise in medical image analysis due to their ability to learn spatial hierarchies of features directly from raw image data, eliminating the need for manual feature extraction [5].

In this research, we propose an AI-based scoliosis screening system that utilizes a CNN model trained on 259 labelled spinal X-ray images [1] to classify conditions as either normal or scoliosis-affected. The model, although constrained by a modest dataset and class imbalance (188 scoliosis, 71 normal), demonstrates the feasibility of deep learning-based scoliosis detection. It is deployed as a Streamlit-based web application that allows clinicians, technicians, and patients to upload spinal X-rays and receive real-time diagnostic

predictions, complementing rather than replacing radiologists.

As illustrated in Fig. 1, the visual difference between a healthy spine and a scoliosis-affected spine underscores the potential of image-based classification for automated screening.

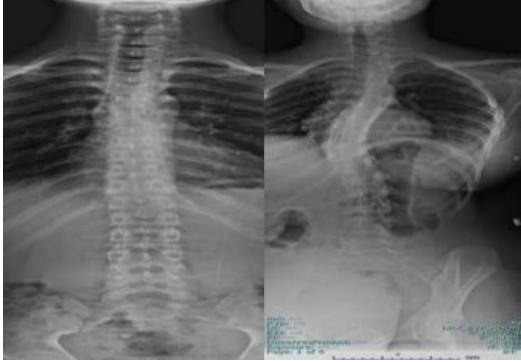


Figure 1: Comparative X-ray showing a normal spine vs. a scoliosis-affected spine.

This work aims to enhance the accuracy, speed, and accessibility of scoliosis diagnosis, particularly in remote or resource-limited settings. The study also sets the foundation for future extensions such as severity grading, explainability (e.g., Grad-CAM visualization), and model integration into telehealth platforms [8], [9].

II. METHODOLOGY

A. Dataset

A publicly available dataset from Mendeley Data was utilized, consisting of 259 labelled spinal X-ray images [1]. Among these, 188 images correspond to scoliosis-affected patients and 71 represent normal individuals. The dataset is specifically curated for scoliosis detection and is suitable for deep learning applications.

While the dataset provides a reliable foundation, its limited size and class imbalance pose constraints on model generalization — an important consideration addressed in Section IV through proposed mitigation strategies.

B. Preprocessing

All images were converted to grayscale and resized to **224×224 pixels**. Pixel intensities were normalized to the range [0, 1], and

categorical labels (Normal / Scoliosis) were encoded for binary classification.

This preprocessing pipeline ensures uniform input dimensions, reduces computational complexity, and enhances model performance by standardizing data distribution [10].

C. CNN Model Architecture

The proposed CNN model comprises stacked convolutional and pooling layers, followed by flattening and fully connected dense layers. Dropout layers were incorporated to minimize overfitting and improve generalization.

The architecture design draws inspiration from prior CNN-based approaches in scoliosis and other medical image classification studies [5], [9].

Model training employed **Binary Cross-Entropy** as the loss function and the **Adam optimizer** with a learning rate of 0.001. Training was conducted for **10 epochs** with a **batch size of 16** and a **validation split of 20%** [10].

D. Evaluation Metrics

The model achieved a **test accuracy of 78.85%** and a **loss of 0.4723** after 10 epochs.

Training and validation accuracy progression across epochs is illustrated in **Fig. 2**, which indicates stable learning and generalization performance.

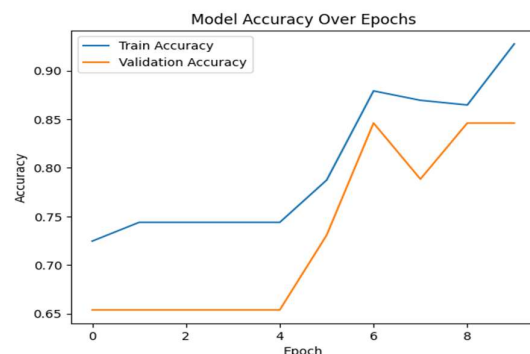


Figure 2. Training and validation accuracy of the CNN model across 10 epochs. The upward trend in both curves suggests consistent model convergence and effective feature learning.

III. SYSTEM DESIGN & IMPLEMENTATION

The scoliosis detection pipeline integrates multiple platforms and frameworks for efficient model training, deployment, and real-time user interaction.

- **Google Colab** was employed for model training, utilizing its cloud-based GPU environment for faster computation on the scoliosis X-ray dataset [1].
- **TensorFlow/Keras** libraries were used to design and train the CNN architecture, providing reliable APIs for deep learning model development [9], [10].
- **PyCharm** served as the local development environment for organizing the web application codebase.
- **Streamlit** was adopted for model deployment, offering a lightweight interface where users can upload spinal X-ray images and instantly receive classification results.

The deployed **Streamlit web application** provides a simple, responsive interface that enables real-time scoliosis screening. By facilitating rapid detection, the system acts as a **decision-support tool** for clinicians rather than replacing radiologists, especially benefiting **rural healthcare centres, schools, and telehealth setups**.

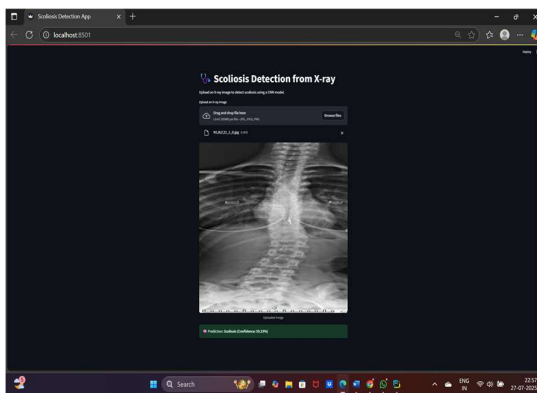


Figure 3. Graphical User Interface (GUI) of the scoliosis detection web application built using Streamlit. Users can upload spinal X-ray images, process them through the CNN model, and obtain real-time diagnostic results.

IV. RESULTS

The proposed scoliosis detection system, developed using a custom Convolutional Neural Network (CNN), was trained and validated on a dataset [1] comprising **188 scoliosis** and **71 normal** spinal X-ray images. The model was designed to classify each X-ray as either Scoliosis or Normal.

A. Model Performance

The CNN model was trained for **10 epochs** using the **Binary Cross-Entropy** loss function and the **Adam optimizer**, following best practices in deep learning for medical imaging [9], [10].

- **Training Accuracy:** 88.65%
- **Validation Accuracy:** 78.85%
- **Validation Loss:** 0.4723

Both training and validation curves demonstrated stable convergence with minimal overfitting.

The performance results confirm that the model effectively captures discriminative spinal curvature features, despite the limited dataset size.

Figure 2 illustrates the training and validation accuracy across epochs, confirming progressive learning and generalization.

B. Graphical User Interface (Streamlit App)

To enhance accessibility and real-time usability, a lightweight **Streamlit web application** was developed [1]. The interface allows users to upload spinal X-ray images in .jpg or .png format and obtain immediate classification results.

Upon image upload, preprocessing is automatically applied, and the model predicts whether the case is Scoliosis or Normal.

The interface is intuitive and requires no technical expertise, making it ideal for use in **rural clinics, schools, and community health programs**.

Figure 3 depicts the Graphical User Interface (GUI) of the Streamlit application showing real-time scoliosis prediction.

C. Confusion Matrix Analysis:

To further evaluate performance, a confusion matrix was generated on the test dataset (Fig. 4). It indicates:

- **True Positives (TP):** Correctly identified scoliosis cases
- **True Negatives (TN):** Correctly identified normal cases
- **False Positives (FP):** Normal images misclassified as scoliosis
- **False Negatives (FN):** Scoliosis images misclassified as normal

The matrix shows that the model performs more confidently in detecting scoliosis than normal cases, likely due to **dataset imbalance (188 vs. 71)** [1], [10].

Future iterations will apply **data augmentation** and **transfer learning** to improve performance on underrepresented normal samples.

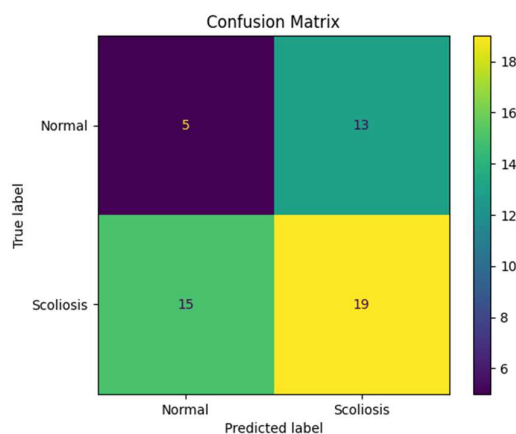


Figure 4. Confusion matrix illustrating the classification performance of the scoliosis detection CNN model.

D. Overall Observations:

- The CNN demonstrated strong accuracy considering the limited dataset [1], [4].
- The Streamlit interface enhances accessibility and real-time screening [1].
- The system complements radiologists by enabling **preliminary screening** and

prioritizing suspected cases for expert review.

- With additional data and clinical validation, the model can evolve into a **decision-support tool** for early scoliosis detection [5], [11].

V. CONCLUSION

This study demonstrates the potential of deep learning, particularly Convolutional Neural Networks (CNNs), for the **early and automated detection of scoliosis** from spinal X-ray images. The proposed model, trained on a publicly available dataset, achieved encouraging performance in distinguishing between normal and scoliotic spines. Its deployment as a **Streamlit-based web application** bridges the gap between advanced AI models and real-world clinical usability.

The system is designed to **support rather than replace radiologists**, enabling faster preliminary assessments and assisting in prioritizing suspected scoliosis cases, especially in **rural or under-resourced regions** where expert access is limited. By offering an intuitive and accessible diagnostic interface, the model demonstrates potential for **telemedicine and community health screening**.

Although the model performed well, limitations remain due to the **small and imbalanced dataset (188 scoliosis vs. 71 normal images)**, which may affect generalization. Future work will focus on addressing this through **data augmentation, transfer learning, and collaborations with medical institutions** for larger, clinically diverse datasets.

Further improvements will include the integration of explainability tools (e.g., Grad-CAM heatmaps) to visualize decision regions, **severity grading of scoliosis**, and possible **3D spine reconstruction from 2D X-rays**. Integration with **Electronic Health Record (EHR) systems** can enhance workflow efficiency and real-world adoption.

Overall, this work underscores the role of **AI-driven diagnostic tools** in improving early detection, reducing diagnostic delays, and

enhancing accessibility in orthopaedic healthcare. With continued development and clinical validation, such systems could contribute significantly to **scalable, non-invasive, and affordable scoliosis screening** across diverse healthcare settings.

VI. LIMITATIONS

Despite the encouraging results achieved through the proposed scoliosis detection system, several limitations were identified during its development and evaluation:

1. Limited Dataset Size and Imbalance:

The dataset comprised only 259 X-ray images (188 scoliosis and 71 normal cases), leading to a noticeable **class imbalance**. This restricts the model's generalization across varied spinal deformities, body postures, and demographic groups. The lack of diverse age and severity cases may introduce **bias in prediction** and limit real-world applicability.

2. Restricted to Binary Classification:

The current model performs **binary classification**—detecting the presence or absence of scoliosis. It does not provide additional clinical insights such as **Cobb angle estimation, curve location (thoracic/lumbar), or severity grading (mild/moderate/severe)**, which are essential for diagnostic and treatment decisions.

3. 2D Imaging Constraint:

As scoliosis is inherently a **three-dimensional spinal deformity**, analysis based solely on **2D X-ray images** limits the model's ability to assess spinal rotation and sagittal-plane curvatures. Incorporating multi-view or 3D imaging modalities could improve diagnostic precision.

4. Lack of Clinical Validation and Compliance:

The system has not yet been **validated in clinical settings** or tested across diverse hospital datasets. For real-world deployment, compliance with **medical data regulations** (e.g., HIPAA, CDSCO, or FDA) and **ethical standards** must be established. Collaborations

with medical professionals will be crucial for future clinical trials.

5. Dependence on Image Quality and User Input:

The Streamlit-based application relies on user-uploaded X-ray images. Variations in **image quality, contrast, or orientation** can affect prediction accuracy. Standardized preprocessing or automated quality-check modules may mitigate this limitation in future iterations.

VII. FUTURE ENHANCEMENTS

1. Dataset Expansion and Augmentation:

Future work will involve curating a **larger and more balanced dataset** that includes diverse age groups, postures, and scoliosis severities. Advanced **data augmentation** and transfer learning techniques will be applied to enhance generalization and reduce overfitting caused by limited data diversity.

2. Multi-Class Classification and Severity Detection:

Beyond binary detection, future versions will enable **multi-class classification** to categorize scoliosis by severity (mild, moderate, severe) using **Cobb angle estimation**. The system could also be extended to identify **curve types** (thoracic, lumbar, double curves), improving its clinical utility for orthopaedic practitioners.

3. 3D Reconstruction and Advanced Imaging Integration:

Since scoliosis is inherently three-dimensional, incorporating **multi-view X-rays** or integrating with **MRI/CT imaging** could enable **3D spine reconstruction**, improving diagnostic accuracy and reducing errors from 2D-only projections.

4. Mobile and Edge Deployment:

The CNN model can be **compressed and optimized** for deployment on **mobile or edge devices**, enabling real-time scoliosis screening in rural and remote areas without requiring high-end computational hardware or continuous internet connectivity.

5. Integration with Clinical Systems:

Integrating the system with **Hospital Information Systems (HIS)** and **Electronic Health Records (EHRs)** will streamline workflows by automatically storing patient data, generating diagnostic reports, and supporting **telemedicine applications** for remote consultations.

6. Explainability and Visualization (Grad-CAM):

To build clinician trust, future iterations will incorporate **explainable AI tools** such as Grad-CAM to visualize regions of interest influencing the model's predictions. This will enhance **model transparency**, support interpretability, and promote acceptance among radiologists and healthcare professionals.

VIII. REFERENCES

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